ARL Spec: Local Weather Monitoring

Brendan Griffiths

Contents

1 Service overview 2
1.1 General state and routines 2
1.2 Message conventions 2
2 Transmitted data 4
2.1 Telemetry 4
2.2 State (retained) 5
2.3 Events 6
3 Controls 6
4 Alerts 7
5 Configuration 8
6 Scenario events 9

1 Service overview

The Local Weather Monitoring (LWM) service is a network of five small weather stations ("sites") spread around the ARL. Topic prefix: `arl/lwm`.

Entity kind	Ids	Notes
`site`	`lwm-1` to `lwm-5`	Weather stations. Each publishes its position (`lat`, `lon`) in `$topology` and as retained state. The sites span about 6 km east to west and 4 km north to south, 3 to 4 km apart.
`service`	`global`	The service as a whole: the lightning summary, the number of active sites, alert summary.

1.1 General state and routines

- **Sampling**: every site samples every 60 s by default. The interval can be set per site from 10 s to 3600 s (multiples of 10 s). Mean wind, maximum gust, rainfall and strike counts aggregate the period since the site's previous sample.
- **Site-specific instruments**: `lwm-3` carries a sonic anemometer, `lwm-4` a soiling station (clean and exposed reference solar cells), `lwm-5` an evaporation pan. Each publishes an extra stream at the site's sample interval.
- **Power**: each site runs on a 12 V battery charged by a small solar panel. Shorter sample intervals use more power, so a site left on a short interval overnight can run its battery flat. A site whose battery is exhausted goes offline until it has recharged.
- **Store and forward**: a site that loses its link buffers up to 720 samples (12 h at 60 s). When the link returns it re-publishes the buffered samples, oldest first, with their original `ts` and `seq` and `backfill: true`, interleaved with new live samples (`backfill: false`). Alerts are judged on live samples only.
- **Calibration rota**: a technician recalibrates each site every 21 to 35 days (`last_calibration` updates). Recalibration can also be requested.
- **Pan refill rota**: every day at 08:00 local a technician tops the `lwm-5` pan up to 205 mm if it is below 190 mm (`last_refill` updates).
- **Lightning**: each site detects strikes within 30 km. Strikes seen by at least one enabled, online site are published as events and summarised once a minute.
- **Technician visits** (`service_anemometer`, `clean_reference_cell`, `refill_pan`) cannot start while lightning is close (Lightning Proximity active) and are abandoned if it becomes active during the visit. A site runs at most one long-running command at a time.
- **Load**: about 0.15 messages/s at the default interval (5 weather, 3 site-specific, 1 lightning summary a minute), plus lightning events during storms; about 0.8 messages/s with every site at 10 s.

1.2 Message conventions

Every service follows the same MQTT layout. `<entity>` is an entity id (or `global` for the whole service).

Topic	Retained	Content
`arl/lwm/$topology`	yes	Machine-readable list of the entities, streams (with field names, types, units and periods), state items, events and controls of the service. Re-published when it changes.
`arl/lwm/<entity>/telemetry/<stream>`	no	Periodic measurements.
`arl/lwm/<entity>/state/<item>`	yes	The current value of a state item, re-published when it changes.
`arl/lwm/events/<kind>`	no	One-off events, including `alert`.
`arl/lwm/cmd/<entity>/<control>`	no	Commands sent by the platform.
`arl/lwm/ack/<command_id>`	no	Acknowledgements of a command.

Telemetry and event payloads carry an envelope: `v` (schema version, integer, currently 1), `ts` (simulated time, Unix milliseconds), `seq` (per-topic sequence number, restarts with the service; gaps reveal lost messages) and `entity` (entity id). The measurement fields are added next to the envelope. A measurement that cannot be taken is `null`. State payloads are `{"v", "ts", "entity", "item", "value"}`.

Commands are JSON `{"command_id": <string>, "researcher": <researcher code>, "value": <value>}`. The researcher must be verified by the ARL API for the service. Acknowledgements carry `command_id`, `target`, `control`, `researcher`, `state` (`accepted`, `rejected`, `in_progress`, `completed`, `failed`), `progress` (0 to 1 for long-running controls), `reason` and `detail`. Rejection and failure reasons: `controls_disabled`, `malformed`, `unknown_control`, `unknown_target`, `unknown_researcher`, `researcher_defunct`, `researcher_not_permitted`, `api_unavailable`, `invalid_value`, `precondition_failed`, `busy` (the target is already running a long command), `fault`, `timeout`, `interrupted`.

Alerts are events on `arl/lwm/events/alert` with `alert_id`, `type` (the alert key below), `severity` (`info`, `warning`, `critical`), `state` (`raised` or `cleared`), `value`, `limit`, `unit`, `message` and `metadata`. An alert that escalates is raised again with the higher severity under the same `alert_id`. Limits come from the ARL API (`/v1/alert-limits/` + `lwm`).

Every service also publishes these:

Measurement	Key	Level	Type / unit	Description
Online	`online`	Global	boolean	Retained state. `true` while the service runs, `false` when it stops or loses its broker connection (last will).
Active alerts	`active_alerts`	Global	array of alert ids	Retained state. The ids of the alerts currently raised, in the order they were raised.

2 Transmitted data

2.1 Telemetry

2.1.1 Weather (`arl/lwm/<site>/telemetry/weather`, every sample interval, default 60 s)

Measurement	Key	Level	Type / unit	Description
Temperature	`temperature_k`	Site	float, kelvin	Air temperature at the site.
Humidity	`humidity_pct`	Site	float, % relative	Relative humidity at the site, 0 to 100.
Pressure	`pressure_hpa`	Site	float, hPa	Atmospheric pressure at the site (station level, about 870 hPa).
Wind speed	`wind_speed_ms`	Site	float, m/s	Mean wind speed over the sample interval.
Wind gust	`wind_gust_ms`	Site	float, m/s	Highest 3 s gust over the sample interval.
Wind direction	`wind_direction_rad`	Site	float, radians	Mean direction the wind comes from, clockwise from true north.
Rainfall	`rainfall_mm`	Site	float, mm	Rain since the previous sample (tipping bucket, multiples of 0.2 mm).
Solar irradiance	`irradiance_wm2`	Site	integer, W/m²	Solar radiation on the site's sensor.
Lightning strikes	`lightning_strikes`	Site	integer, count	Strikes within 30 km of the site since the previous sample.
Battery voltage	`battery_v`	Site	float, volts	Voltage of the site's battery.
Status	`status`	Site	boolean	The site is in service. During a recalibration the site keeps publishing with every measurement `null`.
Backfill	`backfill`	Site	boolean	`true` for a buffered sample re-published after a link outage, `false` for a live sample.
Heartbeat	`ts` (envelope)	Site	timestamp, ms	Time of the latest live message from the site.

2.1.2 Sonic anemometer (`arl/lwm/lwm-3/telemetry/sonic`, every sample interval)

Measurement	Key	Level	Type / unit	Description
Eastward wind	`wind_u_ms`	Site `lwm-3`	float, m/s	Mean air motion towards the east over the interval.
Northward wind	`wind_v_ms`	Site `lwm-3`	float, m/s	Mean air motion towards the north over the interval.
Vertical wind	`wind_w_ms`	Site `lwm-3`	float, m/s	Mean vertical air motion (positive upwards).
Turbulence intensity	`turbulence_intensity`	Site `lwm-3`	float, ratio	Standard deviation of wind speed over its mean. `null` in near calm (mean below 0.5 m/s).
Gust factor	`gust_factor`	Site `lwm-3`	float, ratio	3 s gust over the mean wind. `null` in near calm.
Direction spread	`direction_std_rad`	Site `lwm-3`	float, radians	Standard deviation of the wind direction over the interval.
Sonic temperature	`sonic_temperature_k`	Site `lwm-3`	float, kelvin	Air temperature measured from the speed of sound. Independent of the site's main thermometer.

Measurement	Key	Level	Type / unit	Description
Backfill	`backfill`	Site `lwm-3`	boolean	As for the weather stream.

2.1.3 Soiling station (`arl/lwm/lwm-4/telemetry/soiling`, every sample interval)

Measurement	Key	Level	Type / unit	Description
Reference irradiance	`reference_irradiance_wm2`	Site `lwm-4`	integer, W/m²	Irradiance on a reference cell that is kept clean.
Exposed irradiance	`exposed_irradiance_wm2`	Site `lwm-4`	integer, W/m²	Irradiance on a cell left exposed to dust, like the solar panels.
Soiling ratio	`soiling_ratio`	Site `lwm-4`	float, ratio 0 to 1	Exposed over reference irradiance: the fraction of light that still reaches a dusty panel. Falls on dry days and in dust, rises when rain washes the cell. `null` when the reference is below 50 W/m² (night).
Backfill	`backfill`	Site `lwm-4`	boolean	As for the weather stream.

2.1.4 Evaporation pan (`arl/lwm/lwm-5/telemetry/pan`, every sample interval)

Measurement	Key	Level	Type / unit	Description
Pan level	`pan_level_mm`	Site `lwm-5`	float, mm	Water depth in the Class A evaporation pan (0 to 255 mm; it overflows above 255 mm).
Evaporation	`evaporation_mm`	Site `lwm-5`	float, mm	Evaporation since the previous sample: the level drop corrected for rain. `null` for a sample that contains a refill or an overflow.
Backfill	`backfill`	Site `lwm-5`	boolean	As for the weather stream.

2.1.5 Lightning summary (`arl/lwm/global/telemetry/lightning`, every 60 s)

Measurement	Key	Level	Type / unit	Description
Nearest strike distance	`nearest_strike_km`	Global	float, km	Distance from the ARL centre to the nearest strike in the last minute. `null` if there was none.
Strikes last minute	`strikes_last_min`	Global	integer, count	Strikes detected by any site in the last minute.

2.2 State (retained)

Measurement	Key	Level	Type / unit	Description
Enabled	`enabled`	Site	boolean	Whether the site is enabled.
Sample interval	`sample_interval_s`	Site	integer, seconds	The site's current sample interval.
Last calibration	`last_calibration`	Site	timestamp, ms	Time of the site's last recalibration (rota visit or requested).
Last anemometer service	`last_anemometer_service`	Site	timestamp, ms	Time the cup anemometer was last serviced. `null` if never.
Position	`position`	Site	object `{lat, lon}`, degrees	The site's location (WGS84).

Measurement	Key	Level	Type / unit	Description
Last cleaning	`last_cleaning`	Site `lwm-4`	timestamp, ms	Time the exposed reference cell was last cleaned. `null` if never.
Last refill	`last_refill`	Site `lwm-5`	timestamp, ms	Time the evaporation pan was last topped up. `null` if never.
Active sites	`active_sites`	Global	integer, count	Number of enabled sites currently reporting.

2.3 Events

Measurement	Key	Level	Type / unit	Description
Lightning strike	`arl/lwm/events/lightning`: `distance_km`, `bearing_rad`, `detected_by`	Global	float km, float radians, array of site ids	One event per strike detected by at least one enabled, online site: its distance and bearing (clockwise from north) from the ARL centre, and which sites detected it. `ts` is the strike time.
Alert	`arl/lwm/events/alert`	Global / Site	alert object	Alert raised or cleared (see *Alerts*).

3 Controls

Control	Key	Level	Value (type / unit)	Description
Enable Site	`enable_site`	Site	boolean	Enable or disable the site. A disabled site publishes nothing and does not count as offline.
Set Sample Interval	`sample_interval`	Site	integer, seconds, 10 to 3600, multiple of 10	How often the site samples. The next sample is one new interval after the previous one. Shorter intervals drain the battery faster.
Request Recalibration	`request_recalibration`	Site	boolean, must be `true`	Recalibrates the site's sensors (60 s, `in_progress`). Rejected while the latest gust is above the High Wind warning limit (`sensor_disturbed`) or while the site is replaying buffered samples (`backfill_in_progress`). Fails if the site is offline.
Service Anemometer	`service_anemometer`	Site	boolean, must be `true`	Technician visit to service the cup anemometer: 30 to 120 min `in_progress`, then `last_anemometer_service` updates. Rejected while Lightning Proximity is active.
Clean Reference Cell	`clean_reference_cell`	Site `lwm-4`	boolean, must be `true`	Technician visit to clean the exposed cell: 30 to 120 min `in_progress`, then the soiling ratio returns to 1.0 and `last_cleaning` updates. Rejected while Lightning Proximity is active.
Refill Pan	`refill_pan`	Site `lwm-5`	boolean, must be `true`	Technician visit to top the pan up to 205 mm (5 min). Rejected if the level is above 195 mm (`not_needed`) or while Lightning Proximity is active.

4 Alerts

Alert	Key	Level	Severity, limit key (default, unit)	Description
Site Offline	`site_offline`	Site	warning; no limit	An enabled site has missed 3 live samples. `metadata.reason` is `battery`, `comms` or `no_samples`. Clears on the next live sample.
High Wind	`high_wind`	Site	warning, critical; `lwm.high_wind.warn_ms` (15 m/s), `lwm.high_wind.crit_ms` (20 m/s)	A sample's gust is above the warning or critical limit. Clears after 10 min with the gust below 80% of the warning limit.
Lightning Proximity	`lightning_proximity`	Global	warning; `lwm.lightning.max_km` (10 km)	A strike within the limit of the ARL centre. Clears 30 min after the last such strike.
Storm Approach	`storm_approach`	Global	warning; `lwm.storm.approach_km` (25 km)	At least 3 strikes in the last 10 min whose centre is within the limit and closer than 10 min earlier. `metadata.distance_km` is the nearest recent strike. Clears when no strike has been within the limit for 30 min, or the activity has moved away for 20 min.
Extreme Temperature	`extreme_temperature`	Site	warning; `lwm.temperature.min_k` (268 K), `lwm.temperature.max_k` (313 K)	Temperature outside the limits for 3 samples in a row. Clears 1 K inside the limits.
Heavy Rain	`heavy_rain`	Site	warning; `lwm.rain.max_mm_h` (10 mm/h)	Rainfall over the last hour above the limit. Clears below half the limit.
Low Battery	`low_battery`	Site	warning; `lwm.battery.min_v` (11.5 V)	Battery voltage below the limit. Clears 0.3 V above it.
Sensor Fault	`sensor_fault`	Site	warning; no limit	A measurement is outside its physical range or has not changed for 30 samples (wind ignored in calm, irradiance at night). `metadata.measurement` and `metadata.reason` (`out_of_range`, `unchanged`) say which. Clears when the readings are plausible again, e.g. after recalibration.
Calibration Overdue	`calibration_overdue`	Site	info; `lwm.calibration.max_days` (30 days)	`last_calibration` is older than the limit. Clears on recalibration.
Soiling	`soiling`	Site `lwm-4`	warning; `lwm.soiling.min_ratio` (0.95, ratio)	Soiling ratio below the limit for 3 daylight samples. Clears above the limit plus 0.02 (rain or cleaning).
Pan Level	`pan_level`	Site `lwm-5`	info; `lwm.pan.min_mm` (175 mm), `lwm.pan.max_mm` (230 mm)	Pan level outside the limits. Clears inside them.

5 Configuration

The service reads a TOML file (`config/` + `lwm.toml` in the bundle, `--config` or `ARL_CONFIG` for a binary). Environment variables override the file, and command-line options override both (see *Running the ARL*). Defaults are the values in the provided file.

5.0.1 Common keys

Key	Type / unit	Default	Description
`seed`	integer	`1`	Random seed. The same seed, start time, scenario and commands give the same data. Env `ARL_SEED`.
`speedup_multiplier`	float, ×	`1.0`	Simulation speed relative to real time, up to the service's `max_speedup`. Env `ARL_SPEEDUP`.
`control_enabled`	boolean	`true`	Whether the service accepts commands. When `false` every command is rejected (`controls_disabled`). Env `ARL_CONTROL_ENABLED`.
`start_time`	`"now"` or RFC 3339	`"now"`	Simulated time at start-up. Env `ARL_START_TIME`.
`scenario`	string, path	`""`	Scenario file to load, empty for none. Env `ARL_SCENARIO` (compose file).
`mqtt.address`	string	`"mosquitto"`	MQTT broker host. Env `MQTT_BROKER_ADDRESS`.
`mqtt.port`	integer	`1883`	MQTT broker port. Env `MQTT_BROKER_PORT`.
`mqtt.user`	string	`""`	Broker username, empty for anonymous. Env `MQTT_BROKER_USER`.
`mqtt.password`	string	`""`	Broker password. Env `MQTT_BROKER_PASSWORD`.
`mqtt.queue_capacity`	integer, messages	`200000`	Outgoing messages the service buffers while the broker is slow or unreachable.
`api.address`	string	`"api"`	ARL API host. Env `ARL_API_ADDRESS`.
`api.port`	integer	`8080`	ARL API port. Env `ARL_API_PORT`.
`api.require_researcher`	boolean	`true`	Whether commands must carry a researcher code the API accepts for this service.
`api.researcher_cache_ttl_s`	integer, s	`60`	How long a researcher check is remembered.
`api.alert_limits_refresh_s`	integer, s	`300`	How often the alert limits are re-read from the API.
`health.port`	integer	`9100`	Port of the service's `/healthz` endpoint (used by `healthcheck`). Env `HEALTH_PORT`.

5.0.2 LWM keys (`[lwm]`)

Key	Type / unit	Default	Description
`num_sites`	integer, 1 to 5	`5`	Number of sites, from `lwm-1` upwards. The site-specific instruments exist only if their site does.
`sample_interval_s`	integer, s, 10 to 3600	`60`	Starting sample interval of every site (a multiple of 10). Change it per site with Set Sample Interval.
`lightning_detect_km`	float, km	`30`	Range within which a site detects lightning.
`geojson`	string, path	`"/assets/arl.geojson"`	Site map with the site positions.

Key	Type / unit	Default	Description
`max_speedup`	float, ×	`500`	Highest `speedup_multiplier` the service accepts.
`buffer_samples`	integer, samples	`720`	Samples a site buffers while its link is down.
`backfill_samples_per_tick`	integer, samples	`10`	Buffered samples a site re-publishes per 10 s of simulated time after its link returns.
`initial_soiling_ratio`	float, ratio 0 to 1	`0.97`	Soiling ratio of the `lwm-4` exposed cell at start-up.

6 Scenario events

Events with `service = "lwm"` (format in *Running the ARL*). Every LWM event needs a site as `entity`. Weather (`service = "env"`: storms, dust, heatwaves, cold snaps) affects every site.

Fault	Entity	Params (default)	Effect
`stuck_sensor`	Site	`measurement` (`temperature_k`; also `humidity_pct`, `pressure_hpa`, `wind_speed_ms`, `irradiance_wm2`)	The measurement freezes at its current value (Sensor Fault after 30 samples). Lasts until recalibration or `repair`.
`spike`	Site	`measurement` (as above)	The measurement reports an impossible value (Sensor Fault) for `duration` (default 3 samples).
`comms_loss`	Site	none	The site stops publishing live samples for `duration` (Site Offline), then backfills what it buffered.
`battery_fault`	Site	none	The battery drains much faster for `duration` (Low Battery, possibly Site Offline).
`calibration_age`	Site	`days` (40)	Sets the last calibration to `days` before the event (Calibration Overdue).
`soiling`	`lwm-4`	`ratio` (0.9)	Sets the exposed cell's soiling ratio (Soiling).
`pan_level`	`lwm-5`	`level_mm` (160)	Sets the pan level and pauses the daily refill for `duration` (default 1 day) (Pan Level).
`repair`	Site	none	Ends stuck sensors, spikes, link loss and battery faults on the site.

AI Declaration: Claude Code (Opus 5.5) - generated from the arl-sim source, reviewed